

AVINASH CHHETRI

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AI Engineer | LLM Applications • RAG Systems • Python Backend

PROFESSIONAL SUMMARY

AI Engineer with production Python experience building LLM applications, retrieval-augmented generation systems, and backend automation pipelines. Professionally grounded in enterprise data engineering at TCS. Independently built and evaluated a full-stack RAG pipeline, a fine-tuned domain LLM, and a production-grade backend API. Focus on retrieval system architecture, measurable evaluation, and reliable deployment.

TECHNICAL SKILLS

- **AI & LLM Engineering:** RAG Pipelines, LLM Applications, Semantic Search, NLP, Vector Search, Embedding Models, Cross-Encoder Reranking, Prompt Engineering, Prompt Optimization, LangChain, LlamaIndex, OpenAI API, Hugging Face Transformers, Fine-Tuning (LoRA / QLoRA / PEFT), Ragas Evaluation
- **Backend & Infrastructure:** Python, FastAPI, AsyncIO, REST APIs, API Integration, PostgreSQL, Redis, Docker, Git
- **Cloud & Data:** AWS (Lambda, S3, CloudWatch), Vector Databases (Pinecone, FAISS, ChromaDB), SQL, ETL Pipelines, MLOps Basics

EXPERIENCE

Assistant System Engineer (AI & Backend Engineering)

Tata Consultancy Services | June 2024 – Present

- Backend automation and data engineering for a compliance platform serving global pharmaceutical clients.
- Designed and owned Python automation pipelines for deployment validation and dataset reconciliation, reducing manual operational effort by ~30% across recurring release cycles.
- Wrote and maintained SQL and VQL scripts for validation and transformation of enterprise-scale datasets feeding downstream regulatory reporting.
- Built rule-based watchlist automation that flagged data integrity violations before QA escalation, shrinking the correction feedback loop across engineering and operations.
- Independently diagnosed and resolved production failures — identified root causes, implemented fixes, and coordinated resolution with QA and engineering teams.

PROJECTS

RAG Knowledge Assistant | Python, LangChain, FastAPI, Pinecone, OpenAI API | [GitHub](#)

- Implemented recursive document chunking (512 tokens, 10% overlap) with metadata-aware filtering to prevent cross-document context bleed.
- Benchmarked Pinecone vs. FAISS; selected Pinecone for retrieval consistency and native filtered-search support.
- Added cross-encoder reranking over top-20 vector candidates before LLM context assembly — improved Ragas faithfulness from 0.67 to 0.81 on a 200-query holdout set.
- Final evaluation scores: faithfulness 0.81, answer relevance 0.76, context recall 0.79.
- Exposed ingestion and query endpoints via FastAPI with structured JSON logging and LangSmith tracing; end-to-end query latency averaged under 1.8s.
- Identified and documented consistent degradation on multi-hop queries requiring synthesis across 3+ chunks as a known architectural limitation.

Fine-Tuned Domain Support Assistant | PyTorch, Hugging Face, QLoRA, PEFT | [GitHub](#)

- Fine-tuned Mistral-7B on 4,000 instruction samples using QLoRA (4-bit quantization, LoRA r=16, alpha=32).
- Achieved 22% ROUGE-L improvement over the zero-shot baseline.
- Applied early stopping at epoch 3 after identifying overfitting onset from training loss curves.
- Optimized for single 16GB GPU inference via bitsandbytes quantization at ~2.4s average response latency.

Financial Transaction Backend API | FastAPI, PostgreSQL, Docker | [GitHub](#)

- Built REST API for concurrent wallet operations with ACID-compliant transaction handling and normalized schema design.
- Implemented JWT auth with refresh token rotation, request rate limiting, and structured logging.
- Containerized with Docker for reproducible deployment.

EDUCATION

B.Tech — Computer Science and Engineering | Vellore Institute of Technology | 2020 – 2024